

Quiz 6 Reflection

SCIE 001 Physics

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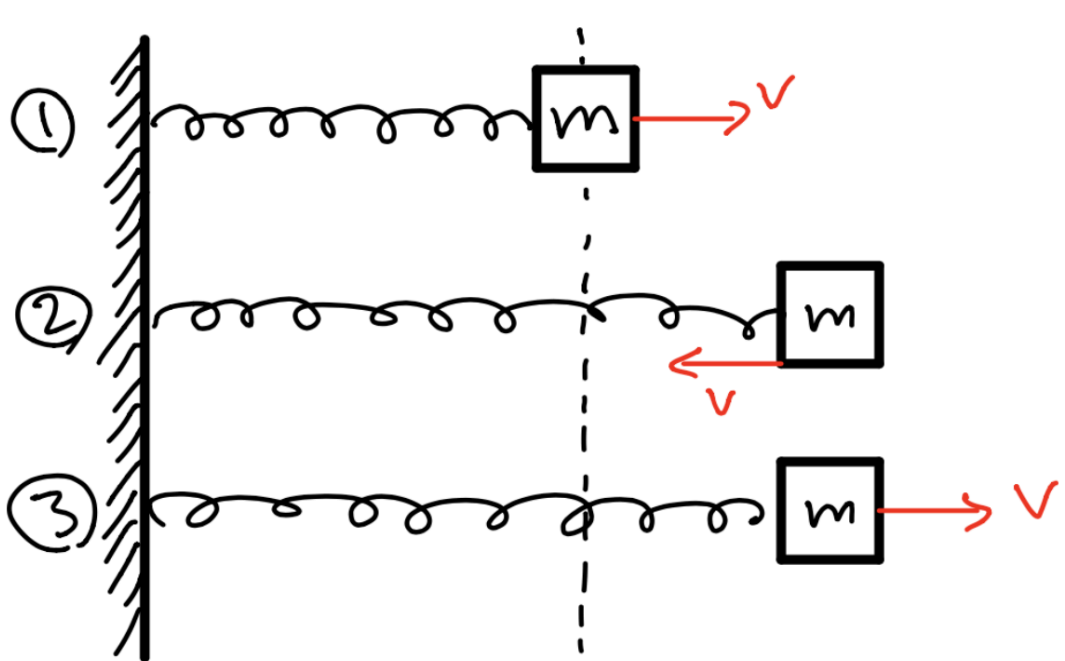
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1. Three Oscillators

1.1. Specify Question

Question 1: Three Oscillators

Consider 3 identical masses attached to 3 identical springs, each with identical speed at the moment shown below. The dashed line shows the equilibrium position and there is no damping. What can you say about the **maximum velocity** attained by each of these oscillators?



☐ $v_3 > v_1 > v_2$

☐ $v_1 > v_2 = v_3$

☐ $v_2 > v_1 > v_3$

☐ $v_2 > v_3 > v_1$

☐ $v_2 = v_3 > v_1$

☐ Not enough information to answer.

☐ $v_1 = v_2 = v_3$

☒ $v_3 > v_2 > v_1$ ✖

✖ 0%

Correct answer

$v_2 = v_3 > v_1$

1.2. Diagnosis Phase

Maximum velocity is directly proportional to maximum amplitude.

$v_{\max} = A\omega$

I misread the diagram and incorrectly thought that the 3rd mass-spring system was stretched further than the second spring in the snapshot.

If correct, the third mass would reach a greater amplitude than the second mass. I thought this because, when the second mass was traveling back outwards, it would be at the same position with opposite velocity, and since I mistook that the third was further outstretched with same velocity, the third would go farther - therefore, having a greater amplitude and greater maximum velocity.

Words: 93

1.3. Correction Phase

I was incorrect in reading that the blocks were at different distances when the snapshot was taken.

Reading the graph properly, there are two ways of thinking about this problem. First with energy, at some stretch x and some velocity v :

$$E_{\text{max amplitude}} = E_{\text{total}}$$
$$E_{\text{spring max amplitude}} + E_{\text{kinetic max amplitude}} = E_{\text{kinetic}} + E_{\text{spring}}$$
$$\frac{1}{2}kA^2 + \frac{1}{2}m(0)^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$
$$A = \sqrt{\frac{mv^2}{k} + x^2}$$
$$v_{\max} = \omega\sqrt{\frac{mv^2}{k} + x^2} \qquad v_{\max} = \omega A$$
$$v_{\max} = \sqrt{\frac{k}{m}}\sqrt{\frac{mv^2}{k} + x^2} \qquad \omega = \sqrt{\frac{k}{m}}$$
$$v_{\max} = \sqrt{v^2 + \frac{kx^2}{m}}$$

$$v_{\max \ 1} = \sqrt{v^2 + \frac{k(0)^2}{m}} = |v|$$
$$v_{\max \ 2} = \sqrt{(-v)^2 + \frac{k(x)^2}{m}} = \sqrt{v^2 + \frac{k(x)^2}{m}} \qquad \text{where } x \text{ is the distance stretched}$$
$$v_{\max \ 3} = \sqrt{v^2 + \frac{k(x)^2}{m}}$$

$$|v| < \sqrt{v^2 + \frac{k(x)^2}{m}} = \sqrt{v^2 + \frac{k(x)^2}{m}}$$
$$\therefore v_{\max \ 1} < v_{\max \ 2} = v_{\max \ 3}$$

The second way is via the velocity function. The velocity over time of all the blocks is periodic and is 90° offset from the position over time. Therefore it is possible to have two masses oscillating with the same amplitude but offset in phase at the same position in time, but one with positive velocity and the other negative.

Words: 97